Anny Dai

AI/ML Engineer

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AI/ML Engineer specializing in deep learning and computer vision, with hands-on experience building end-to-end ML systems from data pipelines to production-scale inference. Strong background in model benchmarking, performance optimization, and deployment on GPU-serving platforms. Seeking AI Engineer or Machine Learning Engineer roles.

Skills

Languages	Python, SQL, C/C++, JavaScript, Bash
ML / AI Libraries	PyTorch, TensorFlow, scikit-learn, OpenCV, NumPy, Pandas, Transformers
Engineering Tools	Docker, Git/GitHub, Databricks, MLflow, NVIDIA Triton, Linux, GCP

Education

Master of Science in Applied Computing (CGPA 4.0/4), *University of Toronto* — September 2024 - December 2025
Artificial Intelligence Focus — Toronto, ON, Canada

Honors Bachelor of Science with High Distinction (CGPA 3.94/4), *University of Toronto* — September 2018 - April 2024
Specialist in Computer Science, Major in Mathematics, Minor in Statistics — Toronto, ON, Canada

Technical Experience

Data Science Intern | PyTorch, OpenCV, XGBoost, Databricks, Docker, NVIDIA Triton — **May 2025 – December 2025**
Magna New Mobility — *Toronto, ON, Canada*

- For ultra-high-resolution industrial images where full-image inference was impractical, built a **patch-based segmentation pipeline** that split inputs into windows and stitched predictions, enabling **production-scale inference**.
- To improve detection of small and subtle surface defects, applied targeted data augmentation, model tuning, and optimized stitching strategies, achieving **+0.19 IoU**, **+0.30 Precision**, and **+0.22 Recall** across defect classes.
- For leather defect analysis requiring interpretability and flexibility, used a **hybrid CV–ML workflow** (**binary segmentation → defect crop extraction → XGBoost severity classification**), producing modular and interpretable predictions.
- To support end-to-end deployment and iteration, built **Databricks-based ETL pipelines** and deployed **GPU inference with Docker and NVIDIA Triton**, reducing setup time and enabling reproducible training and deployment workflows.

Software Engineer Intern | C++, Python, ROS2 Foxy, OpenCV, Linux, Git — **May 2022 - June 2023**
Magna Electronics — *Brampton, ON, Canada*

- For **vision-based sensing systems** requiring accurate perception, performed intrinsic and extrinsic camera calibration to ensure reliable sensor alignment and measurement accuracy.
- To improve robustness of autonomous sensing software, resolved bugs and extended **C++/Python modules**, strengthening system stability and reliability.
- To support customer releases and platform integration, managed **version-controlled deployments** for customer delivery and implemented **ROS2 Foxy C++ features on Linux**, while maintaining code quality via testing, reviews, and GitHub workflows.

Projects

Artwork Restoration with Generative Image Inpainting | Python, PyTorch, git — **September 2024 - December 2024**

- Designed and executed a controlled benchmarking study of **3 generative inpainting models** (CO-MOD-GAN, ICT, LaMa) on **WikiArt**, training on **20k paintings** and evaluating on a **5k-image test set** with **5 degradation masks**.
- Quantitatively compared restoration quality using **MSE, MS-SSIM, and FID**, identifying clear trade-offs: LaMa achieved strongest **pixel and structural fidelity** (MSE 0.01002, MS-SSIM 0.79205), while Co-Mod-GAN produced the most realistic global distributions (**FID 112.155**).
- Performed qualitative error analysis across multiple art styles to surface **practical failure modes**, including facial distortion, fine-detail loss, and breakdowns under full occlusion.

INTR: Interpretable Transformer for Image Classification | Python, PyTorch, git — **September 2024 - December 2024**

- Conducted large-scale evaluation of **INTR** on **14 fine-grained datasets** (8 original + 6 extended), combining **accuracy benchmarking** with **attention heatmap analysis**.
- Analyzed generalization behavior by comparing INTR against **ResNet-50**, observing up to **+9.2% accuracy gain** on Fish and up to **-12% degradation** on CUB under complex background conditions.
- Investigated **class-specific cross-attention patterns** to assess interpretability, demonstrating INTR's ability to localize discriminative regions driving fine-grained predictions.

3D Object Tracking & Motion Forecasting | Python, PyTorch, git — **December 2021 - April 2022**

- Developed **LiDAR-based vehicle detection, tracking, and motion forecasting pipelines** for autonomous driving perception, under the supervision of **Prof. Raquel Urtasun**.
- Explored robustness improvements in dynamic scenes through **hard target mining**, **Gaussian target representations**, and **GNN-based modeling**, evaluating their impact on tracking stability and forecasting accuracy.